## ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 63 | One part $= 9$ |
| 2 | **(c)** 24 | Subtract the added number |
| 3 | **(c)** 18 L | $\frac{3}{10} \times 60$ |
| 4 | **(c)** 15 hours | More pipes, less time |
| 5 | **(d)** 55 | A : B : C = 4 : 6 : 5, so one part is 11 |
| 6 | **(d)** 2 km | 1 km = 100000 cm |
| 7 | **(b)** 10 m/s | Multiply by $\frac{5}{18}$ |
| 8 | **(a)** 48 | One part $= 6$ |
| 9 | **(d)** 12 | Subtract the added number |
| 10 | **(b)** 24 L | $\frac{2}{7} \times 84$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | One part $= 98 \div 14 = 7$ (1); numbers 35 and 63 (1). | 1+1 |
| 12 | 45 km/h $= 45 \times \frac{1000}{3600}$ (1) $= 12.5$ m/s (1). | 1+1 |
| 13 | Actual distance $= 6 \times 50000$ cm $= 300000$ cm (1); $= 3$ km (1). | 1+1 |
| 14 | One part $= 90 \div 15 = 6$ (1); numbers 24 and 66 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Milk 35 L, water 5 L (1); new water $= \frac{3}{7} \times 35 = 15$ L (1); add 10 L (1). | 1+1+1 |
| 16 | A : B : C $= 3 : 6 : 5$ (1); one part $= 10$ (1); numbers 30, 60, 50 (1). | 1+1+1 |
| 17 | One part $= 6$; daughter 18 (1); 4 years ago: 38 and 14 (1); ratio $19 : 7$ (1). | 1+1+1 |
| | **OR:** Milk 36 L, water 24 L (1); new water $= \frac{4}{3} \times 36 = 48$ L (1); add 24 L (1). | 1+1+1 |
| 18 | A : B : C $= 6 : 8 : 5$ (1); one part $= 7$ (1); numbers 42, 56, 35 (1). | 1+1+1 |
| | **OR:** One part $= 8$; daughter 16 (1); 6 years ago: 34 and 10 (1); ratio $17 : 5$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Total work $= 12 \times 15 = 180$ man-days (1); (i) $180 \div 18 = 10$ days (2); (ii) $180 \div 9 = 20$ men (2). | 1+2+2 |
| | **OR:** Ratio $3 : 5$ (1); one part ₹ 100 (1); Anil ₹ 300 (1.5); Bhanu ₹ 500 (1.5). | 1+1+1½+1½ |
| 20 | Total work $= 20 \times 24 = 480$ man-days (1); (i) $480 \div 16 = 30$ days (2); (ii) $480 \div 30 = 16$ men (2). | 1+2+2 |
| | **OR:** Ratio $2 : 3$ (1); one part ₹ 300 (1); Anil ₹ 600 (1.5); Bhanu ₹ 900 (1.5). | 1+1+1½+1½ |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

## ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 6 hours | More pipes, less time |
| 2 | **(b)** 45 | A : B : C = 2 : 4 : 3, so one part is 15 |
| 3 | **(d)** 6 km | 1 km = 100000 cm |
| 4 | **(c)** 15 m/s | Multiply by $\frac{5}{18}$ |
| 5 | **(a)** 70 | One part $= 7$ |
| 6 | **(c)** 20 | Subtract the added number |
| 7 | **(d)** 50 L | $\frac{5}{9} \times 90$ |
| 8 | **(c)** 4 hours | More pipes, less time |
| 9 | **(d)** 70 | A : B : C = 6 : 8 : 7, so one part is 10 |
| 10 | **(d)** 7 km | 1 km = 100000 cm |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | 63 km/h $= 63 \times \frac{1000}{3600}$ (1) $= 17.5$ m/s (1). | 1+1 |
| 12 | Actual distance $= 4 \times 200000$ cm $= 800000$ cm (1); $= 8$ km (1). | 1+1 |
| 13 | One part $= 75 \div 15 = 5$ (1); numbers 35 and 40 (1). | 1+1 |
| 14 | 81 km/h $= 81 \times \frac{1000}{3600}$ (1) $= 22.5$ m/s (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Milk 20 L, water 25 L (1); new water $= \frac{8}{4} \times 20 = 40$ L (1); add 15 L (1). | 1+1+1 |
| 16 | A : B : C $= 6 : 9 : 4$ (1); one part $= 13$ (1); numbers 78, 117, 52 (1). | 1+1+1 |
| 17 | One part $= 5$; daughter 20 (1); 5 years ago: 40 and 15 (1); ratio $8 : 3$ (1). | 1+1+1 |
| | **OR:** Milk 45 L, water 27 L (1); new water $= \frac{6}{5} \times 45 = 54$ L (1); add 27 L (1). | 1+1+1 |
| 18 | A : B : C $= 3 : 9 : 7$ (1); one part $= 10$ (1); numbers 30, 90, 70 (1). | 1+1+1 |
| | **OR:** One part $= 4$; daughter 12 (1); 2 years ago: 30 and 10 (1); ratio $3 : 1$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Total work $= 9 \times 40 = 360$ man-days (1); (i) $360 \div 15 = 24$ days (2); (ii) $360 \div 12 = 30$ men (2). | 1+2+2 |
| | **OR:** Ratio $2 : 3$ (1); one part ₹ 400 (1); Anil ₹ 800 (1.5); Bhanu ₹ 1200 (1.5). | 1+1+1½+1½ |
| 20 | Total work $= 14 \times 18 = 252$ man-days (1); (i) $252 \div 21 = 12$ days (2); (ii) $252 \div 12 = 21$ men (2). | 1+2+2 |
| | **OR:** Ratio $2 : 3$ (1); one part ₹ 350 (1); Anil ₹ 700 (1.5); Bhanu ₹ 1050 (1.5). | 1+1+1½+1½ |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 20 m/s | Multiply by $\frac{5}{18}$ |
| 2 | **(d)** 44 | One part $= 4$ |
| 3 | **(b)** 35 | Subtract the added number |
| 4 | **(c)** 42 L | $\frac{7}{12} \times 72$ |
| 5 | **(a)** 12 hours | More pipes, less time |
| 6 | **(b)** 21 | A : B : C = 4 : 10 : 3, so one part is 7 |
| 7 | **(b)** 2 km | 1 km = 100000 cm |
| 8 | **(a)** 25 m/s | Multiply by $\frac{5}{18}$ |
| 9 | **(a)** 45 | One part $= 5$ |
| 10 | **(c)** 12 | Subtract the added number |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Actual distance $= 9 \times 100000$ cm $= 900000$ cm (1); $= 9$ km (1). | 1+1 |
| 12 | One part $= 64 \div 16 = 4$ (1); numbers 12 and 52 (1). | 1+1 |
| 13 | 126 km/h $= 126 \times \frac{1000}{3600}$ (1) $= 35$ m/s (1). | 1+1 |
| 14 | Actual distance $= 5 \times 250000$ cm $= 1250000$ cm (1); $= 12.5$ km (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Milk 30 L, water 36 L (1); new water $= \frac{9}{5} \times 30 = 54$ L (1); add 18 L (1). | 1+1+1 |
| 16 | A : B : C $= 8 : 10 : 3$ (1); one part $= 8$ (1); numbers 64, 80, 24 (1). | 1+1+1 |
| 17 | One part $= 7$; daughter 7 (1); 7 years ago: 35 and 0 (1); ratio $1 : 0$ (1). | 1+1+1 |
| | **OR:** Milk 60 L, water 20 L (1); new water $= \frac{2}{3} \times 60 = 40$ L (1); add 20 L (1). | 1+1+1 |
| 18 | A : B : C $= 4 : 10 : 9$ (1); one part $= 8$ (1); numbers 32, 80, 72 (1). | 1+1+1 |
| | **OR:** One part $= 9$; daughter 9 (1); 3 years ago: 33 and 6 (1); ratio $11 : 2$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Total work $= 16 \times 21 = 336$ man-days (1); (i) $336 \div 12 = 28$ days (2); (ii) $336 \div 14 = 24$ men (2). | 1+2+2 |
| | **OR:** Ratio $2 : 3$ (1); one part ₹ 240 (1); Anil ₹ 480 (1.5); Bhanu ₹ 720 (1.5). | 1+1+1½+1½ |
| 20 | Total work $= 25 \times 12 = 300$ man-days (1); (i) $300 \div 20 = 15$ days (2); (ii) $300 \div 15 = 20$ men (2). | 1+2+2 |
| | **OR:** Ratio $3 : 5$ (1); one part ₹ 120 (1); Anil ₹ 360 (1.5); Bhanu ₹ 600 (1.5). | 1+1+1½+1½ |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.